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Development and Successful Field Trial of Novel Downhole Gas Handling Concept

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Abstract

After development of a high-performance gas separation technology, further research focused on the enhancement of gas handling capabilities. The resulting concept significantly improves gas handling capabilities in the pump stages, mitigates gas locking, reduces the upthrust of the impeller, and boosts gas separator efficiency. A high-performance slug mitigation system, which incorporates this advanced anti-lock gas handling pump concept, was installed for field trials. The findings of these trials will be presented.

This innovative technology prevents the initiation of gas locking at the impeller inlet. A small hole is introduced in the flow path between the high-pressure fluid on the outer side of the rotating impeller and the suction side of the inlet vane of the impeller. Higher pressure fluid flushes away the gas buildup at the inlet edge of the impeller, thereby decreasing gas locking. Additionally, it reduces upthrust created in the high flow, low density fluid conditions. In addition to the increasing anti-gas locking capabilities, it also allows for a higher gas volume fraction (GVF) over a wider operating range. Introducing this concept with a slug mitigation system further increases the slug handling of gas separators and pumps.

This groundbreaking pump stage technology, in conjunction with a high-performance slug mitigation system, was built and tested in the field. Test results showed successful operation under slugging conditions demonstrated by reduced pump intake pressure (PIP), higher drawdown up to 200 PSI, increased gas production and higher gas-liquid-ratio (GLR) handling capabilities. Gas slug mitigation was increased seven times compared to standard gas separators.

The novel leak path at the suction side of the impeller inlet prevents gas buildup by continuously flushing the gas, which causes gas locking. This increases the handling capabilities across a wider operating range. Reduced gas recycling and additional liquid during gas slugging enhance gas handling capabilities. The slug mitigation system provides additional fluid reservoirs during gas slugging for the separator to process and deliver to the pump.

Background

Traditional mechanical gas separators for conventional vertical wells have existed since the mid-1970s. At that time, applications did not have high GLR, and wells were not drawn down significantly. Over time, as operators sought to fully deplete the reservoirs, wells were drawn down further, leading to lower pressure at the pump intake and higher GLR. These wells experience gas bubbles suspended in the liquid stream in separated, turbulent, or clustered bubble flow patterns as the fluid flows upstream to the ESP. The development of new high performance gas separators has almost doubled the flow capabilities of the mechanical gas separators (from 5400 to 10000 BPD in 400 series), and increased gas separation efficiency up to 99%.

Unconventional, horizontal wells with advanced fracturing techniques tend to produce a slugging flow pattern. In this pattern, the fluid flow contains cap bubbles and forms single gas slugs, also known as Taylor bubbles. They create extended periods where only pockets of gas reach the separator intake. Therefore, the mechanical separator has little to zero liquid to process and deliver to the pump.

Production from unconventional horizontal slugging wells varies rapidly from 97% liquid phase to 97% gas phase. The two phases follow a stratified flow pattern in the well's horizontal section, with liquid at the bottom and gas at the top. In the transition zone, the wellbore shifts from horizontal to vertical, causing the stratified gas to break out of the liquid and rise, resulting in a slug flow pattern where high concentrations of gas follow high concentrations of liquid, as seen in Figure 1. Additionally, horizontal wells have undulations or toe-up configurations, creating dissimilar extended-length gas slugs. When this slug flow passes through a tight annulus, the reduced cross-sectional area between the motor and the casing increases the severity of the slug as it elongates. These situations cause significant changes in the gas volume fraction at the intake of the electrical submersible pump (ESP) within short time intervals.

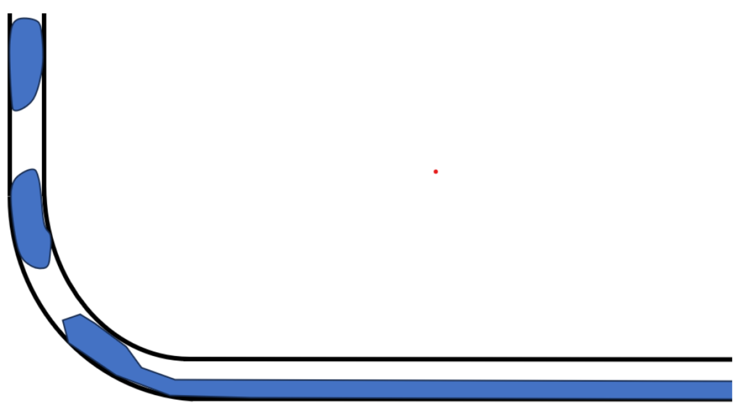

Figure 1—Creation of slug flow at transition from horizontal to vertical (Brown et al., 2024)

In unconventional horizontal wellbores, the hydrodynamics of two-phase flow patterns differ significantly from vertical wells due to the transition zone. The behavior of the two phases in the wells is influenced by the liquid's specific gravity and the gas's buoyancy within a transition zone, which creates the slug flow pattern, whose size and length vary with the formation output of the different phases (Brown et al., 2024). They can intensify with changes in inclination upstream of the shift from horizontal to vertical. During extreme slug flow conditions, conventional gas separators struggle to process and deliver liquid to the pump, causing the stages to gas lock.

Gas interference and locking significantly affect pump performance, causing short-run life, low production, and energy inefficiency. It is the number one cause of ESP failure according to data gathered from customers and historical records. Gas also reduces ESP reliability due to increased shutdowns, start-ups, and motor heating, as gaseous fluid does not provide effective motor cooling compared to liquid (Nicholson et al., 2023; Brown et al., 2023).

As a well is drawn down, the pressure drop at the pump's intake amplifies the gas slugging problems. Operators have employed several conventional gas handling methods and techniques to avoid, separate, or handle the gas with limited success. Various methods and techniques for avoiding gas entry into the ESPs include inverted shrouds, passive separators, mechanical gas separators, and gas handling pumps (Sheth et al., 1996, Wilson et al., 2003). Most wells lack the diametrical clearance to use inverted shrouds, and passive (reverse flow) separators are only effective at very low liquid rates. Combined natural annular gas separation and conventional mechanical gas separation are insufficient to overcome large slugs because the mechanical separator runs out of liquid to process during an extended gas slug. Special gas-handling pumps have limited gas-handling capabilities.

Slug Mitigation Development

Mechanical gas separators use centrifugal force or vortex generation to separate gas from the well fluid and deliver the fluid volume required by the pump. The remaining extra well fluid exits to the annulus. If any liquid exists in the exit fluid, it returns to the gas separator's intake, and gas vents upwards due to density difference. The gas separators easily manage small gas slugs and help provide liquid-rich fluid to the pump. Returned fluid from the exit recycles and helps provide more liquid-rich fluid to the pump. Large gas slugs can easily fill the separation chamber, which starves the pump of the required liquid-rich well fluid and causes gas lock.

A state-of-the-art transparent test system was developed to understand various flow patterns and gas separator performance. The system allows for unprecedented visualization of mechanical separation phenomena. The gas slugging phenomenon became evident during horizontal to vertical testing in the new test system, as shown in Figure 2. A slugging flow pattern was set up on the test system to study the slugging flow effects on both single and tandem separators. The separator easily ingested small slugs of gas as it maintained adequate phases to process. However, as the gas slugs increased in length and duration, the separator only had gas to process, which was then ingested by the pump.

Figure 2—New two-phase transparent test system

The test system provided evidence of issues with mechanical gas separators in severe gas slugging applications, where they failed because they could not provide a liquid-rich gaseous mixture to the pump. Additionally, conventional mechanical gas separators have limited flow capabilities and experience reduced performance at higher total flow rates and GVFs. Until now, research on these limitations of gas separation technology and improvement to existing gas separator designs has not kept up with the proliferation of horizontal well applications.

During a gas slug event, the velocity of the fluids within a traditional-length separator dispels or process the liquid in a matter of few seconds. The concept of providing a liquid reservoir within the separator was derived, built, and evaluated. Adding length and internal reservoirs within the separator would provide extended time and surplus liquid to process during a gas slugging event. When tested, the internal chambers proved to be an advantage in processing extended slug flow regimes. A field trial prototype was built with eleven internal 13-inch reservoirs between fluid mover stages. Figure 3 depicts the difference in length of a traditional tandem gas separator compared to the slugger prototype.

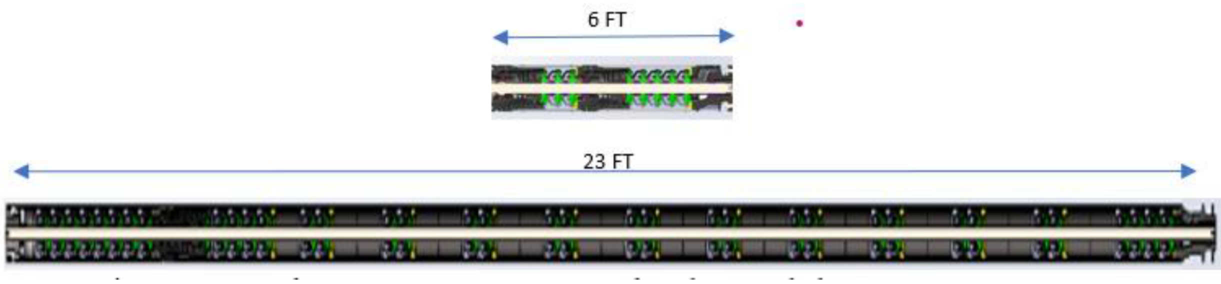

Figure 3—A tandem gas separator compared to the novel slug mitigation system concept

Leak Path Impeller Technology Development

Centrifugal impellers in the pump are designed for incompressible liquid duty with backward-swept impeller vanes. These impeller vanes have a pressure and suction surface. When gaseous fluid enters the impeller, the lowest pressure is near the suction side of the vane at the shroud. Due to the lower density of gas at this point, it cannot move forward which leads to a buildup that blocks the flow passage. This blockage further accumulates along the suction edge of the vane and then moves towards the pressure side, eventually completely obstructing the vane passage. This phenomenon is known as gas locking, as shown in Figures 4A to 4D (Shahid et al., 2021).

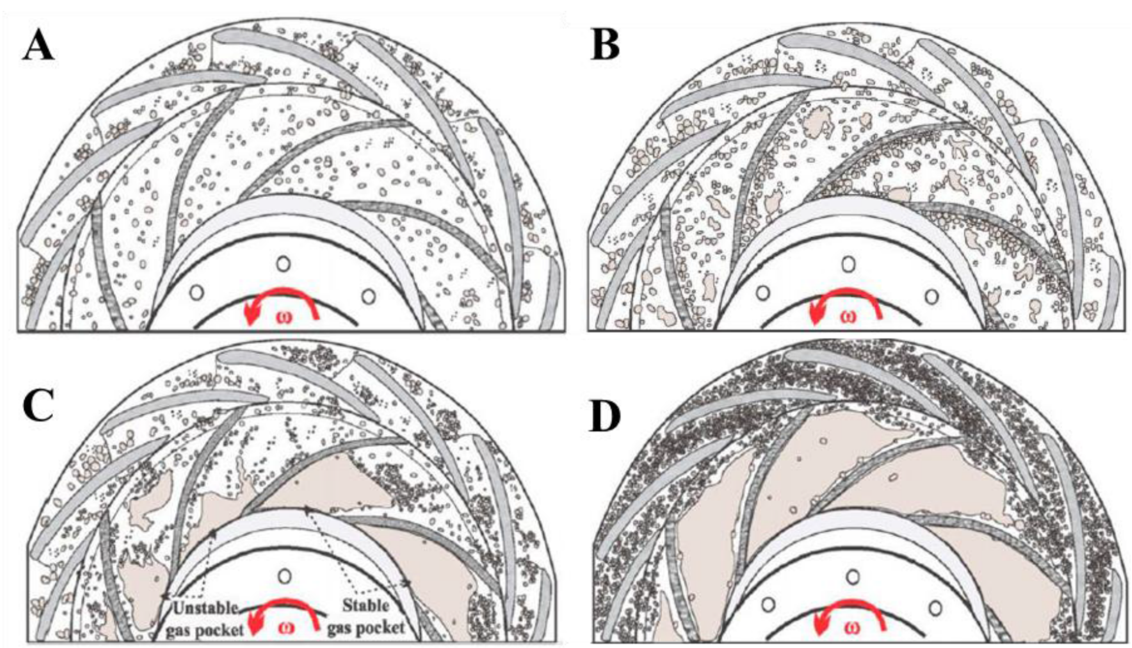

Figure 4—(A) Gas bubbles start building up at the corner of the leading edge of the suction side of the vane and shroud. (B)Gas bubbles start building up along the suction side of the vane. (C) as bubbles start progressing from the suction side to the pressure side of the next vane. (D) Impeller is gas locked. (Shahid at al., 2021).

When impellers are either gas-locked or filled with high gassy fluid, their performance deteriorates due to higher two-phase flow losses and the ineffective transfer of kinetic energy to the gassy fluids. This reduces the pressure head generated by the impeller. As the lift provided by the successive stages diminishes in handling gassy fluid, the fluid lacks sufficient energy to reach the surface.

When the pump lifts fluid, variations in fluid composition create fluctuations in motor loading and amperage. These fluctuations, combined with uneven cooling from the slugging fluid mixture, cause internal heat to rise in the motor, which leads to shutdowns and production loss. Additionally, pump bearings use lubrication and experience additional heat rise, resulting in premature failures. Consequently, ESP reliability is reduced.

A novel concept of inserting a hole drilled in the shroud near the suction side of each vane at the inlet was conceived, as seen in Figure 5. The hole creates a path for high-pressure fluid, recirculating from the impeller exit to the inlet, to enter the hole and continuously flush away the gas bubble built up at the lowest-pressure point. This continuous flushing prevents the gas locking. This is a proprietary, protected technology.

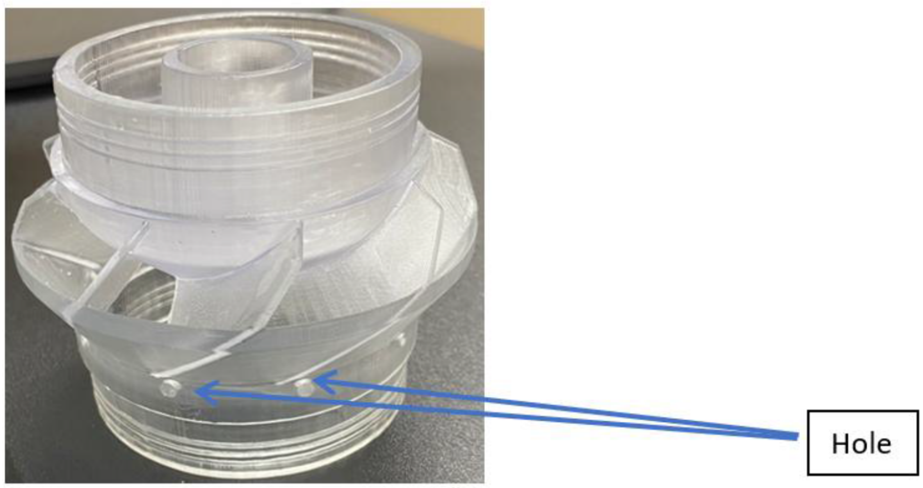

Figure 5—Anti Gas Lock Impeller.

Limited tests under controlled lab conditions showed significant improvement in the gas handling characteristics of the impeller with the anti-gas locking mechanism. The impeller managed higher gas percentage at best efficiency point (BEP) flow rates, and performance deterioration at higher and lower flow rates was reduced.

Slug Mitigation Gas Separator Field Trials and Discussion

Special consideration for field trials was crucial to validate the concept. Specifically, a Permian Basin well was selected where a new high-performance tandem gas separator faced challenges due to extended gas slugs. The well was drilled with a severe toe-up profile and originally operated for years with gas lift due to gas slugging. A new operator wanted to try an ESP in the well, specifically using the new high-performance separator. Early operation of the well with an ESP and a tandem high-performance separator revealed substantial gas slugging events with 43-minute-long liquid low flow or no flow events.

After a brief initial run and a draw-down of about 150 psi, the novel concept of high-performance slug mitigation system was proposed to mitigate the effects of gas slugging. The recommended system was installed and given special attention to its operation, given its novelty and the need to understand its performance. The system responded well to lower casing pressure and, after some adjustments, began to operate smoothly. Daily oil and gas production increased by approximately 73%. Figure 6 illustrates the performance difference between the tandem high-performance separator and the slug mitigation separator.

As shown in Figure 6, the pump intake pressure significantly reduced after the slug mitigation installation, which demonstrated the capability of the system to better process two-phase fluid, reduce gas-only slugs, increase production, and ensure smoother operation of the ESP.

Figure 6—Tandem gas separator performance Vs. Extended length high-performance separator improved drawdown from near 1400 psi with the tandem separator to 700 psi using the extended system.

The concept was also evaluated in steam-assisted gravity drainage (SAGD) wells (Field trial case 2), which required the system to be positioned in near-horizontal areas. These wells also needed an additional gas avoidance system, using a gravity-induced rotating intake to draw fluids from the low side of the casing. Figure 7 shows the before and after installation results and highlights a significant improvement in system stability.

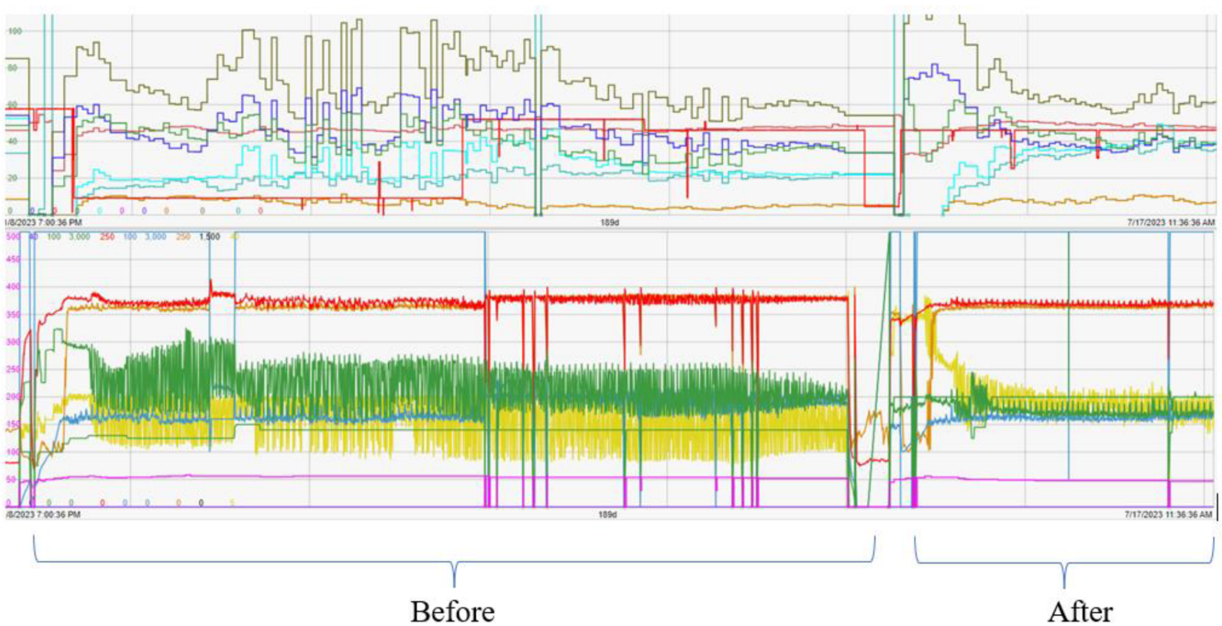

Figure 7—PIP swings differences (in yellow) before and after the installation of the novel slug mitigation system.

To date, 18 additional slug protection systems have been installed in the SAGD market. These systems, with a diameter of 5.38 inches, include a high-performance separator, an extended slug protection system, and a rotating intake. Positive feedback from operators includes the report of significant increases in production and performance.

Leak Path Impeller Field Trial and Discussion

Field trial information for the leak path technology is limited due to the lack of opportunities for pump-only change comparisons. However, some prototypes installed at the end of last year are still operational, enabling the monitoring of improved performance in gassy wells used for field testing. Figures 8 and 9 display the before and after of a field trial well which had a gas separator with a taper-type pump system. Figure 8 displays the performance of the equipment operated in proportional integral derivative (PID) mode due to high presence of gas. The well experienced a high-temperature spike that caused the sensor to fail, affecting the PIP measurement, which was about 2000 psi. Figure 8 also shows two variables erratically swinging: the motor current between 25 and 40 A and the motor voltage between 2500 and 3000 V.

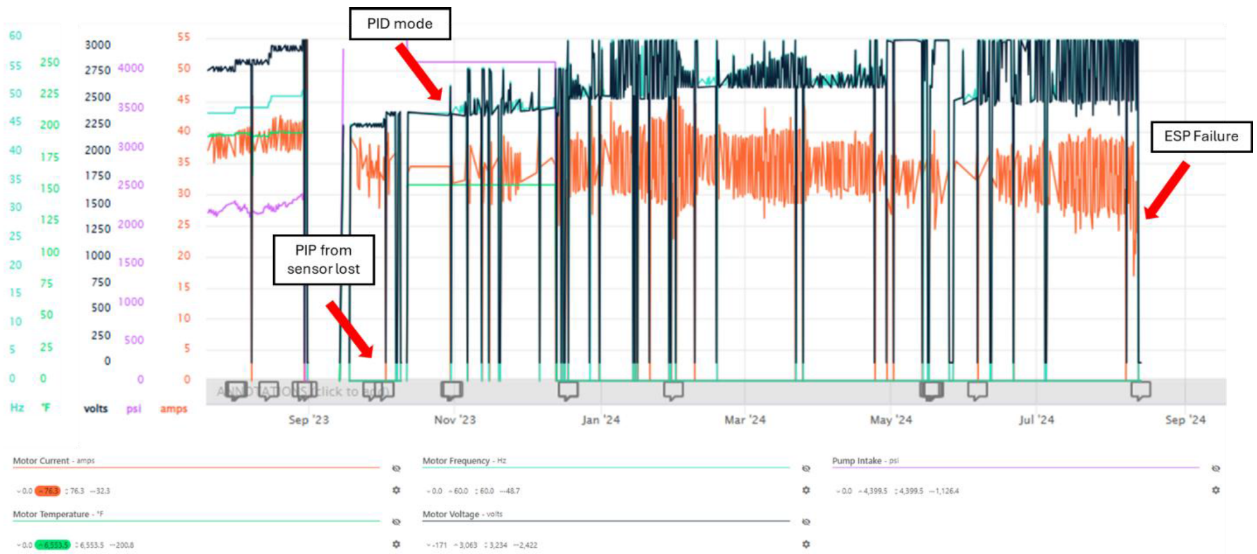

Figure 8—ESP with gas separator and taper-type system ESP.

After the ESP failed, a new ESP was run into the well, using a gas separator with a gas handling pump with leak path impellers (with a capacity of 10000 BPD), and the same producing pumps as the previous run. As shown in Figure 9, the well has been operating with steadier performance for more than 200 days and has maintained a mostly constant PIP of around 750 psi, approximately 1250 psi lower than the previous run. Motor current swings have diminished to a range of between 32 to 37 A, and the voltage is sustained at about 2250 V without PID mode. Hence, the improvements in the monitored variables denote the enhanced performance achieved by using a gas handling pump with leak path impellers, a technology that allows recirculation to prevent gas blockage in the impellers.

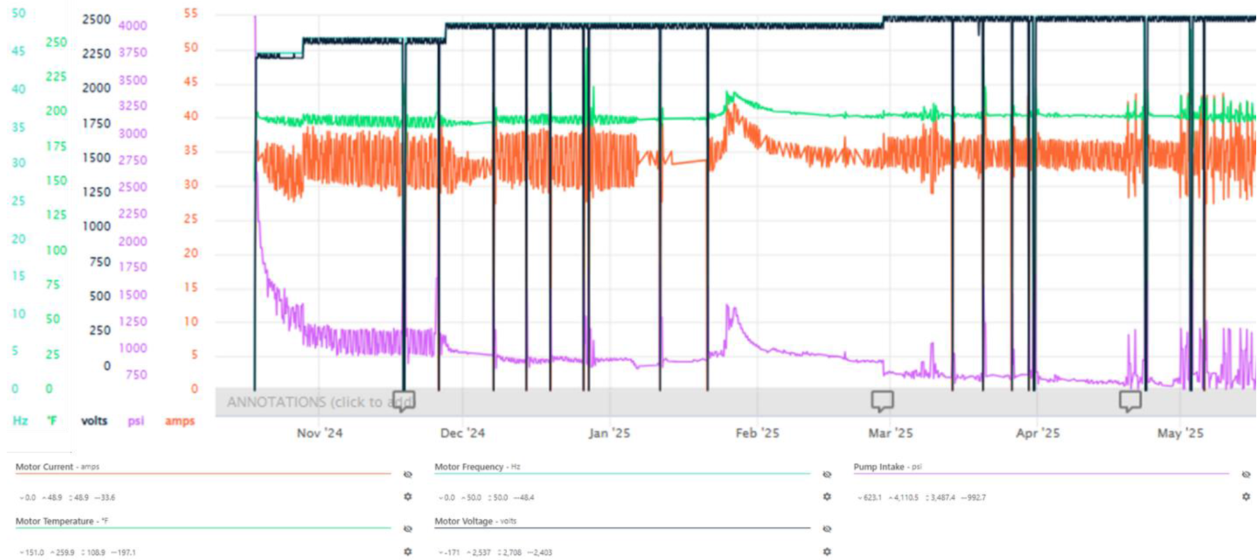

Figure 9—ESP with gas separator and gas handling pump with leak path impellers.

Conclusion

The field trials demonstrated that the combination of high-performance gas separators and novel slug mitigation technology effectively mitigates gas slugging in unconventional wells. This innovative approach has led to several notable performance improvements, including ESPs operating at lower pump intake pressures and in higher gassy conditions, further reduced intake pressures which yields higher production rates, and minimized intake pressure variations which stabilizes production. Additionally, motor performance and run life have been enhanced by maintaining cooler internal temperatures and steady operating conditions, while ESP shutdowns were significantly reduced, increasing overall ESP run life and reliability.

When integrated, high-performance gas handling pumps featuring a novel gas handling stage with slug mitigation systems and high-performance gas separators are expected to substantially extend the operating life of ESPs in gassy and slugging wells. This combination not only boosts gas and oil production rates but also allows the ESP to achieve greater drawdown compared to conventional systems. The promising results from these field trials indicate a significant advancement in the efficiency and reliability of ESPs in challenging well conditions.

The lab studies of the novel leak path impeller indicated a marked increase in gas handling ability over a conventional gas handling stage. For this reason, the design is being incorporated into high performance gas separators and slug mitigation systems as well as multiple gas handling tapered pump designs and in various impeller flow rates. Major benefits are expected in situations where the ESP is expected to ingest 100% of the gas, such as operating below a production packer.

Thanks to ongoing field trials, notable improvement can be observed in the behaviors of the ESP through the reduction of PIP and stabilization of variables like the motor current and motor voltage. Combining these new novel technologies has proven to be effective in high GLR conditions.

Nomenclature

Amp- Electric current (Amp)
GLR- Gas liquid ratio
GVF- Gas volume fraction
ESP- Electrical submersible pump
PIP- Pump intake pressure (psi)

SAGD- Steam-assisted gravity drainage
BEP- Best Efficiency Point
PID- Proportional-Integral-Derivative

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